



Database Management Systems

SQL overview, Data definition,
Structure of SQL queries

Anchor: Dr. Nagasundari S, Dept of CSE

Database Management Systems

Unit 1: Introduction to Database Management and SQL

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Dr. Nagasundari S

Department of Computer Science and Engineering



Alter table actions include:

- Adding or dropping a column (attribute)
- Changing a column definition
- Adding or dropping table constraints

Example:

```
ALTER TABLE people ADD COLUMN Job VARCHAR(12);
```

1) ADD a column in the table

```
ALTER TABLE table_name ADD new_column_name column_definition  
[ FIRST | AFTER column_name ];
```

2) Add multiple columns in the table

```
ALTER TABLE table_name ADD new_column_name column_definition  
[ FIRST | AFTER column_name ], ADD new_column_name column_definition  
[ FIRST | AFTER column_name ],
```

3) MODIFY column in the table

```
ALTER TABLE table_name MODIFY column_name column_definition  
[ FIRST | AFTER column_name ];
```

4) DROP column in table

```
ALTER TABLE table_name  
DROP COLUMN column_name;
```

5) RENAME column in table

```
ALTER TABLE table_name CHANGE COLUMN old_name new_name column_definition  
[ FIRST | AFTER column_name ]
```

6) RENAME table

```
ALTER TABLE table_name  
RENAME TO new_table_name;
```

7) Altering (Changing) a Column Definition or a Name

To change a column's definition, use **MODIFY** or **CHANGE** clause along with the ALTER command. For example, to change column **col** from CHAR(1) to CHAR(10), you can use the following command

```
ALTER TABLE table_name MODIFY col CHAR(10);  
ALTER TABLE table_name MODIFY j BIGINT NOT NULL DEFAULT 100;  
ALTER TABLE table_name ALTER i SET DEFAULT 1000;
```

**Note: i, j are columns*

You can change the column position of MySQL table without losing data with the help of ALTER TABLE command

Syntax:

```
ALTER TABLE yourTableName MODIFY yourColumnName1 data type AFTER /BEFORE  
yourColumnName2;
```

Changing Column's Default Value

Use SET DEFAULT to specify the default value explicitly, or DROP DEFAULT to remove the current default and allow MySQL to assign the “default default.”

To change a default value, use

Syntax:

```
ALTER col_name SET DEFAULT
```

Ex-

```
ALTER TABLE mytbl ALTER j SET DEFAULT 1000;
```

To drop a default value, use

Syntax:

```
ALTER col_name DROP DEFAULT
```

Ex-

```
ALTER TABLE mytbl ALTER j DROP DEFAULT;
```

- We use the alter table command to add attributes to an existing table.
- All tuples in the table are assigned null as the value for the new attribute.
- The form of the alter table command is

ALTER TABLE R ADD A D;

- where r is the name of an existing table, A is the name of the attribute to be added, and D is the type of the added attribute.

- We can drop attributes from a table by the command

alter table r drop A;

- where r is the name of an existing table.
- A is the name of an attribute of the table.
- Many database systems do not support dropping of attributes, although they will allow an entire table to be dropped.

Adding and Dropping Constraints

Change constraints specified on a table

Add or drop a named constraint

Adding constraint

```
ALTER table people add constraint people_gender_fk foreign key (gender)
references gender_tab;
```

Dropping constraint

```
ALTER TABLE Persons DROP CONSTRAINT PK_Person;
```

To drop a column

- Choose either CASCADE or RESTRICT
- CASCADE would drop the column from views etc.
- RESTRICT is possible if no views refer to it.

ALTER TABLE EMPLOYEE DROP COLUMN Address CASCADE;

Default values can be dropped and altered :

ALTER TABLE DEPARTMENT ALTER COLUMN Mgr_ssn DROP DEFAULT;

**ALTER TABLE DEPARTMENT ALTER COLUMN Mgr_ssn SET DEFAULT
'333445555';**

Drop:

- To remove a table from an SQL database, we use the drop table command.
- The drop table command deletes all information about the dropped table from the database
`drop table r; // Complete table structure will be dropped`

Syntax:

```
DROP TABLE table1;
```

```
DROP TABLE IF EXISTS table1;
```

```
DROP TABLE IF EXISTS table1, table2, table3;
```

Truncate:

A truncate SQL statement is used to remove all rows (complete data) from a table. It is similar to the DELETE statement with no WHERE clause.

Syntax:

```
TRUNCATE TABLE table_name;
```

// delete complete data from an existing table. Table Structure remains

Difference between DROP and TRUNCATE in SQL

S.No	DROP	TRUNCATE
1.	It is used to eliminate the whole database from the table.	It is used to eliminate the tuples from the table.
2.	Integrity constraints get removed in the DROP command.	Integrity constraint doesn't get removed in the Truncate command.
3.	The structure of the table does not exist.	The structure of the table exists.
4.	Here the table is free from memory.	Here, the table is not free from memory.
5.	It is slow as compared to the TRUNCATE command.	It is fast as compared to the DROP command.

RENAME:

RENAME TABLE offers more flexibility. It allows renaming multiple tables in one statement.

This can be useful when replacing a table with a new pre-populated version:

Syntax:

```
RENAME table table_name to new table_name
```

```
RENAME TABLE products TO products_old, products_new TO products;
```

Consider 2 tables - 'employee' and 'department'

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(50)  
);
```

```
CREATE TABLE department (  
    id SMALLINT PRIMARY KEY  
);
```

1. Adding a New Column

```
ALTER TABLE employee  
ADD COLUMN department_id INT;
```

2. Reposition a Column ('department_id' after 'name')

```
ALTER TABLE employee  
MODIFY COLUMN department_id INT AFTER name;  
--We use the 'AFTER' keyword to reposition col1 after any col2
```

To place a column as the first column in the table, we use the 'FIRST' keyword

```
ALTER TABLE employee  
MODIFY COLUMN department_id INT FIRST;
```

3. Rename a Column (department_id to dept_ref_id)

```
ALTER TABLE employee  
RENAME COLUMN department_id TO dept_ref_id;
```

4. Change Datatype of a Column (INT to SMALLINT for dept_ref_id)

```
ALTER TABLE employee  
MODIFY COLUMN dept_ref_id SMALLINT;
```

5. Adding Constraints

a). NOT NULL

ALTER TABLE employee

MODIFY COLUMN dept_ref_id SMALLINT **NOT NULL**;

--This ensures that dept_ref_id can never have a null value

b). DEFAULT

ALTER TABLE employee

ALTER COLUMN dept_ref_id SMALLINT NOT NULL **SET DEFAULT** 100;

--This sets the default dept_id as 100 if no value has been assigned to it

c). CHECK

ALTER TABLE employee

ADD CONSTRAINT chk_dept **CHECK** (dept_ref_id > 0);

--The chk_dept is a check constraint which will check if the value inserted is >0 and if not, it returns an error.

d). PRIMARY KEY

Suppose we want to make dept_ref_id as the primary key instead of emp_id. For this, we first have to drop the existing primary key (emp_id) and then add the primary key constraint for dept_ref_id

ALTER TABLE employee

DROP PRIMARY KEY;

ALTER TABLE employee

ADD CONSTRAINT pk_emp **PRIMARY KEY** (dept_ref_id);

e). FOREIGN KEY

- ON DELETE CASCADE, ON UPDATE CASCADE

```
ALTER TABLE employee  
ADD CONSTRAINT fk_dept  
FOREIGN KEY (dept_ref_id) REFERENCES department(id)  
ON DELETE CASCADE  
ON UPDATE CASCADE;
```

If the foreign key constraint is being added to an attribute which already has an existing foreign key constraint, we first have to drop the existing constraint and then add a new one. Otherwise, it is not necessary.

```
ALTER TABLE employee  
DROP FOREIGN KEY fk_dept; -- In this case, the foreign key is named fk_dept
```

- ON DELETE SET NULL

```
ALTER TABLE employee  
ADD CONSTRAINT fk_null  
FOREIGN KEY (dept_ref_id) REFERENCES department(id)  
ON DELETE SET NULL;
```

- ON DELETE SET DEFAULT
-- Set column default before applying
ALTER TABLE employee
MODIFY COLUMN dept_ref_id SMALLINT **DEFAULT 1;**

```
ALTER TABLE employee  
ADD CONSTRAINT fk_default  
FOREIGN KEY (dept_ref_id) REFERENCES department(id)  
ON DELETE SET DEFAULT;
```

e). Dropping Constraints

- DROPPING DEFAULT

```
ALTER TABLE employee  
ALTER COLUMN dept_ref_id DROP DEFAULT;
```

- DROPPING CHECK

```
ALTER TABLE employee  
DROP CONSTRAINT chk_dept;
```

- DROPPING PRIMARY KEY

```
ALTER TABLE employee  
DROP PRIMARY KEY;
```

- DROPPING FOREIGN KEY

```
ALTER TABLE employee  
DROP FOREIGN KEY fk_default;
```

f). Dropping a Column

```
ALTER TABLE employee  
DROP COLUMN dept_ref_id;
```

1. Which command is used to delete all records from a table without removing its structure?

- A) DELETE
- B) DROP
- C) TRUNCATE
- D) ALTER

2. In SQL, what does the DESCRIBE command do?

- A) Shows the details of a table structure
- B) Deletes a table
- C) Modifies a table
- D) Creates a new schema

3. Which of the following statements is used to remove an entire table from the database?

- A) DELETE
- B) DROP
- C) TRUNCATE
- D) REMOVE

4. The ALTER TABLE statement is used to?

- A) Remove a table
- B) Modify an existing table structure
- C) Retrieve table data
- D) Define a new schema

Answers:

1. C
2. A
3. B
4. B



THANK YOU

Anchor: Dr. Nagasundari S, Dept of CSE

nagasundaris@pes.edu